BUSINESS INTELLIGENCE · TECHNICAL DOCUMENTATION

Customer Value & Risk Analytics Dashboard

Churn prediction model, Power BI data model, segmentation logic, measure reference and report specification for a customer churn and portfolio analytics solution.

Author	Leprince Dongmo Nanda
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> **How to read this document.** Sections 1 to 9 describe the solution exactly as it is built today. Section 10 states what the solution does not do, and section 12 lists planned work that is not delivered. Every figure quoted was read from the notebook output, the source files or the `.pbix` itself.

1 Purpose and scope

The solution has two parts. A **machine-learning pipeline** (Python, scikit-learn) estimates the churn risk of bank customers and is served through a Streamlit application. A **Power BI report** gives a retention and portfolio view of a retail banking customer base: how many customers are at risk of leaving, how that risk distributes across relationship managers and cities, and how customer value is spread across the book.

1.1 Business questions answered by the report

Question	Answered by
How large is the customer base, and how much activity does it generate?	KPI cards: TotalCustomers, TotalAccounts, TotalTransactions
What is the overall churn exposure of the portfolio?	AvgChurnProbability, CustomersAtRisk
Is churn risk concentrated on particular relationship managers?	Churn by relationship manager chart
Where is churn risk concentrated geographically?	Churn probability by city map
How much income does the segment under review represent?	Annual income gauge and average income card
Does tenure differ by occupation?	Tenure by occupation chart

1.2 Intended audience

- Retention and CRM teams: identify at-risk customers and the segments they sit in.
- Portfolio and relationship managers: compare churn exposure across managed books.
- Management: a single overview of base size, activity and risk.

1.3 Out of scope

- The churn score is consumed, not computed, in Power BI (see §2.2).
- Revenue, fee and margin analysis: no revenue field exists in the source.
- Time-series and cohort analysis (§10.3), row-level security (§10.5).

2 Data sources

2.1 Source files

File	Rows	Grain	Content	Used by
`Customer-Churn-Records.csv`	10,000	One row per customer	Public, anonymised Bank Customer Churn dataset (Kaggle), 18 columns	Model training
`customers.csv`	200	One row per customer	Demographics, income, tenure, acquisition channel, city, relationship manager, churn probability	Power BI
`accounts.csv`	407	One row per account	Account type, current balance, credit score, owning customer	Power BI
`transactions.csv`	8,758	One row per transaction	Transaction type, amount, date, originating account	Power BI
`products.csv`	309	One row per customer–product holding	Product name and subscription date per customer	Power BI

The four Power BI files are **synthetic datasets**, generated to simulate the relational structure of a retail bank's operational systems. They contain no real customer data. `customers.csv` and `transactions.csv` use a semicolon delimiter and a decimal comma; `accounts.csv` and `products.csv` use a comma delimiter.

2.2 Origin of the churn probability shown in the dashboard

The `ChurnProbability` column of `customers.csv` is **not calculated inside Power BI and is not an output of the model described in §3**. It is a simulated score (drawn from a Beta-shaped distribution, mean 0.30, maximum 0.65) that was generated together with the synthetic customer table.

> **Consequence for interpretation.** The Random Forest is trained on the Kaggle dataset, whose columns (credit score, balance, number of products…) do not exist for the synthetic customers, so the model cannot score them. The churn figures in the report therefore demonstrate the reporting mechanism — how a scored attribute flows into a decision-support layer — and not a validated risk estimate. In a production setting the simulated column would be replaced by the model output, joined on the customer key.

2.3 Refresh

The report uses import mode against local CSV files. There is no scheduled refresh and no gateway: the model is refreshed manually in Power BI Desktop when a source file changes.

3 The churn prediction model

This section records how the model in the notebook `CHURN MODELING_REPORT.ipynb` is built, how well it performs, and what it can and cannot support.

3.1 Training dataset

Property	Value
Rows × columns	10,000 × 18
Missing values	None. Every column is fully populated.
Geography	Three countries: France (5,014 customers), Spain, Germany
Card type	Four levels, most frequent DIAMOND (2,507)
Target	`churn`, binary
Class balance	20.4% churners. The classes are imbalanced.

> **Why the class balance matters.** Predicting "no churn" for every customer would already score 79.6% accuracy. Accuracy alone therefore says very little, and the model must be judged on the churn class (§3.5).

3.2 Leakage and identifier removal

The correlation scan against the target returned `Complain` at r = 0.996, effectively a copy of the label: a customer is recorded as having complained almost exactly when they churn. The column was dropped before training. Had it been kept, every metric in §3.5 would have been near-perfect and meaningless.

The identifier columns `RowNumber`, `CustomerId` and `Surname` are also removed from the features. The real `CustomerId` is kept aside and written to the exported score file.

After removal, the remaining linear signals are weak: Age (r = 0.285), IsActiveMember (r = −0.156), Geography (r = 0.154), Balance (r = 0.119). This is the genuine difficulty of the problem and explains the recall ceiling below.

3.3 Features and preprocessing

Thirteen features reach the models: `CreditScore, Geography, Gender, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember, EstimatedSalary, Satisfaction Score, Card Type, Point Earned`. Preprocessing is held inside a scikit-learn `ColumnTransformer` nested in a `Pipeline`, so every transformation is fitted on the training fold only and applied identically at scoring time.

- Numeric features are standardised with `StandardScaler`.
- Categorical features (`Geography`, `Gender`, `Card Type`) are one-hot encoded with `drop="first"`.

3.4 Models and selection

Model	Configuration
Logistic Regression	`max_iter=10000`, default regularisation. Interpretable baseline.
Random Forest	`n_estimators=300, max_depth=None, random_state=42`

The data is split 80/20 with `stratify=y` and `random_state=42`: 8,000 training and 2,000 test rows (408 churners in the test set). Library version: scikit-learn 1.6.1.

3.5 Performance on the held-out test set

Metric	Logistic Regression	Random Forest
Accuracy	0.813	0.869
Precision (churn class)	0.62	0.80
Recall (churn class)	0.21	0.47
F1 (churn class)	0.32	0.59
ROC-AUC	0.778	0.867
Churners found, of 408	86	193
False positives, of 1,592 non-churners	52 (3.3%)	48 (3.0%)

Random Forest was selected. It more than doubles recall on the churn class, from 0.21 to 0.47, while raising precision from 0.62 to 0.80. Logistic regression collapses onto the majority class: it reaches 0.813 accuracy mainly by predicting "no churn", barely above the 0.796 trivial baseline.

- **Precision 0.80.** When the model flags a customer, it is right four times out of five.
- **Recall 0.47.** The model finds slightly under half of the customers who leave: 215 of the 408 churners in the test set were missed. It is a prioritisation tool, not an exhaustive early-warning system.
- The 86.9% accuracy figure should not be quoted on its own: it is a gain of 7.3 points over the trivial baseline.

The most important features of the Random Forest (impurity-based) are Age (0.21), NumOfProducts (0.12), Balance (0.11), EstimatedSalary (0.11) and CreditScore (0.10).

3.6 Risk bands

Predicted probabilities are bucketed into four bands by a single function, `risk_level()`, shared by the notebook and the Streamlit application.

Band	Probability range	Test customers	Share
Low risk	0.00 to 0.20	1,249	62.4%
Moderate	0.20 to 0.40	405	20.2%
High risk	0.40 to 0.70	222	11.1%
Critical	0.70 to 1.00	124	6.2%

3.7 Deliverables

Artefact	Content
`churn_model.pkl`	The selected Random Forest pipeline (preprocessing + classifier), saved with joblib under scikit-learn 1.6.1.
`customer_churn_predictions.csv` / `.xlsx`	2,000 test customers: real `CustomerId`, `Churn_Probability`, `Risk_Level`.
Streamlit application	Loads `churn_model.pkl`, scores an uploaded CSV with the 13 feature columns and returns probability and risk band. `scikit-learn` is pinned to 1.6.1 in `requirements.txt`.

3.8 Limitations of the score

- **The probabilities are not calibrated.** A Random Forest output of 0.70 is a ranking position rather than a 70% likelihood. The ordering is reliable; the absolute value is not.
- **Scores cover the test set only.** Probabilities were exported for the 2,000 held-out rows, not for all 10,000 customers.
- **No hyperparameter tuning and no resampling.** Neither a grid search nor a class-rebalancing technique is applied. Both are realistic levers for raising recall and are recorded in the roadmap.
- **The model does not feed the dashboard.** See §2.2.

4 Data model

4.1 Diagram

Customers	Accounts	Transactions	Products
CustomerID (PK) Age, Gender, City, Region Occupation, AnnualIncome TenureYears, AcquisitionChannel RelationshipManager ChurnProbability Segment (calculated)	AccountID (PK) CustomerID (FK) AccountType CurrentBalance CreditScore	TransactionID (PK) AccountID (FK) TransactionType Amount Date	CustomerID (FK) ProductName SubscriptionDate (FK to Date)

Figure 1. Tables of the Power BI model. A fifth table, `Date` (one row per day, 2020 to 2025), is related to Products.

4.2 Relationships

From (one)	To (many)	Cross-filter	Active
Customers[CustomerID]	Accounts[CustomerID]	Single	Yes
Accounts[AccountID]	Transactions[AccountID]	Single	Yes
Customers[CustomerID]	Products[CustomerID]	Single	Yes
Date[Dates]	Products[SubscriptionDate]	Single	Yes

Key integrity was checked on the source files: every account belongs to a known customer, every transaction to a known account, and every product holding to a known customer.

4.3 Design note: this is a chain, not a pure star

In a textbook star schema every dimension joins directly to the fact table. Here Transactions joins to Customers only through Accounts, which makes the customer, account and transaction path a snowflake chain. Accounts plays a double role: a dimension when analysed by account type or credit score, and a bridge when transaction measures are sliced by customer attributes. The design is acceptable at this data volume; filters from Customers reach Transactions across two hops.

5 Data dictionary

5.1 Customers

Column	Type	Definition
CustomerID	Text	Unique customer identifier (CUST0001 to CUST0200). Primary key.
Age	Whole number	Age in years (18 to 74).
Gender	Text	Male, Female or Other.
City	Text	One of 20 Belgian cities. Geography key of the map visual.
Region	Text	Belgian province or region of the city (9 values, for example Hainaut, Antwerp, Brussels-Capital).
Occupation	Text	Engineer, Worker, Manager, Retired, Doctor, Student or Teacher.
AnnualIncome	Whole number	Annual income in euro (20,404 to 198,998).
TenureYears	Decimal	Length of the banking relationship, in years (0.3 to 19.9).
AcquisitionChannel	Text	Branch, Online, Referral or Call Center.
RelationshipManager	Text	Assigned manager code RM1 to RM4, or Unassigned (49 customers). See §6.2.
ChurnProbability	Decimal (0–1)	Simulated churn score on a 0 to 1 scale. See §2.2.
Segment	Text (calculated)	Value and risk segment derived in DAX. See §7.

5.2 Accounts

Column	Type	Definition
AccountID	Text	Unique account identifier. Primary key.
CustomerID	Text	Owning customer. Foreign key to Customers.
AccountType	Text	Checking, Savings, Credit or Investment.
CurrentBalance	Decimal	Account balance at extraction date.
CreditScore	Decimal	Credit score on a 300–850 scale.

5.3 Transactions

Column	Type	Definition
TransactionID	Text	Unique transaction identifier. Primary key.
AccountID	Text	Originating account. Foreign key to Accounts.
TransactionType	Text	Deposit, Withdrawal, Payment, Transfer or Investment.
Amount	Decimal	Transaction amount in euro.
Date	Date	Transaction date (15 November 2024 to 14 November 2025). Not used by any visual, see §10.3.

5.4 Products

Column	Type	Definition
CustomerID	Text	Holding customer. Foreign key to Customers.
ProductName	Text	Mortgage, Personal Loan, Credit Card, Mutual Fund or ETF.
SubscriptionDate	Date/time	Date the customer took up the product.

The Products table has no single-column primary key: the natural key is the pair CustomerID + ProductName.

6 Transformations (Power Query)

6.1 Steps applied

Query	Steps
customers	Read CSV (delimiter `;`, UTF-8), promote headers, set explicit types, replace empty `RelationshipManager` with `Unassigned`, rename the source header `Rgion` to `Region`.
accounts	Read CSV, promote headers, type `CurrentBalance` and `CreditScore` as numbers.
transactions	Read CSV (delimiter `;`), promote headers, set explicit types (`Amount` number, `Date` date).
products	Read CSV, promote headers, set explicit types, sort by `SubscriptionDate`.
Date	Generated in M: one row per day from 1 January 2020 to 31 December 2025, with calculated Year, Month, Quarter and Weekend columns.

6.2 Handling the missing relationship manager

49 of the 200 customers have no relationship manager in the source. The empty value is replaced with the literal `Unassigned`. Dropping the rows would bias every customer-level total; leaving nulls would make the unmanaged population disappear from some groupings. Keeping it as its own category lets managed and unmanaged customers be compared directly: the churn-by-manager chart shows Unassigned as a fifth bar alongside RM1 to RM4.

7 Customer segmentation logic

`Segment` is a calculated column on the Customers table. It classifies every customer into exactly one of four categories using a first-match rule.

```
Segment =
SWITCH(
    TRUE(),
    Customers[AnnualIncome] > 100000 && Customers[TenureYears] > 5, "High Value",
    Customers[AnnualIncome] > 50000  && Customers[TenureYears] > 2, "Mid Value",
    Customers[ChurnProbability] > 0.6,                              "At Risk",
    "Other"
)
```

Segment	Condition	Customers
High Value	Annual income above €100,000 and tenure above 5 years	78
Mid Value	Annual income above €50,000 and tenure above 2 years, and not already High Value	75
At Risk	Churn probability above 0.6, and neither High nor Mid Value	1
Other	Everything else	46

> **Known behaviour: the order of the rules is not neutral.** Because `SWITCH(TRUE(), …)` returns on first match, a customer who is both high income and above the 0.6 churn threshold is labelled High Value, never At Risk. Seven customers are above the threshold, but only one carries the At Risk label. The `CustomersAtRisk` measure counts the full at-risk population independently of the segment and should be used for that question.

A calculated column is used rather than a measure because the label must be evaluated once per customer and then used in a slicer.

8 Measure reference

Eleven measures are defined in the model. Six are used on the report page; five are defined but not placed on any visual. Three further figures on the page use implicit aggregations taken directly from a column.

8.1 Measures used on the report page

`TotalCustomers`

```
TotalCustomers = DISTINCTCOUNT(Customers[CustomerID])
```

`TotalAccounts`

```
TotalAccounts = DISTINCTCOUNT(Accounts[AccountID])
```

`TotalTransactions`

```
TotalTransactions = COUNTROWS(Transactions)
```

AvgChurnProbability

```
AvgChurnProbability = AVERAGE(Customers[ChurnProbability])
```

Unweighted mean per customer: a large and a small customer count equally.

CustomersAtRisk

```
CustomersAtRisk =
CALCULATE(DISTINCTCOUNT(Customers[CustomerID]), Customers[ChurnProbability] > 0.6)
```

The 0.6 threshold is hard-coded; changing it means editing the measure.

AvgChurn_By_RM

```
AvgChurn_By_RM =
CALCULATE(AVERAGE(Customers[ChurnProbability]),
          ALLEXCEPT(Customers, Customers[RelationshipManager]))
```

> **Known behaviour: this measure ignores the segment slicer.** `ALLEXCEPT` clears every filter on Customers except the one on RelationshipManager. The manager chart therefore always shows each manager's churn across their whole book (RM1 0.33, RM2 0.32, RM3 0.30, Unassigned 0.29, RM4 0.25), even when a segment is selected.

8.2 Measures defined but not used on the page

Measure	Expression
CustomersByRM	`DISTINCTCOUNT(Customers[CustomerID])`
CustomersWithProduct	`DISTINCTCOUNT(Products[CustomerID])`
ProductPenetration	`DIVIDE(DISTINCTCOUNT(Products[CustomerID]), DISTINCTCOUNT(Customers[CustomerID]))`
TotalDepositAmount	`CALCULATE(SUM(Transactions[Amount]), Transactions[TransactionType] = "Deposit")`
AvgTransactionValue	`DIVIDE(SUM(Transactions[Amount]), COUNTROWS(Transactions))`

The model also contains an empty placeholder measure named `Mesure 2`, which should be deleted.

8.3 Implicit aggregations

Figure on the page	Field	Aggregation
Income gauge	Customers[AnnualIncome]	Sum
Average income card	Customers[AnnualIncome]	Average
Tenure by occupation chart	Customers[TenureYears]	Sum

9 Report specification

The report has one page, sized 1280 × 720 pixels (16:9).

Visual	Type	Field / measure and purpose
Report title	Text box	"Customer Value & Risk Analytics Dashboard"
Accounts	Card	TotalAccounts: size of the account base
Churn probability	Card	AvgChurnProbability: headline risk level
Transactions	Card	TotalTransactions: activity volume
Customers	Card	TotalCustomers: size of the base
Customers at risk	Card	CustomersAtRisk: count above the 0.6 threshold
Average income	Card	Average of AnnualIncome
Total income	Gauge	Sum of AnnualIncome
Segment	Slicer	Customers[Segment]: filters the page, with the exception noted in §8.1
Churn by manager	Combo chart	AvgChurn_By_RM by RelationshipManager
Tenure by occupation	Combo chart	Sum of TenureYears by Occupation
Risk by city	Map	AvgChurnProbability by City, bubble size by risk

9.1 Reference figures

Figure	Whole base (no selection)	"High Value" selected
TotalCustomers	200	78
TotalAccounts	407	162
TotalTransactions	8,758	3,497
AvgChurnProbability	0.30	0.33
CustomersAtRisk	7	3
Average annual income	€109.72K	€147.49K

> Every card reflects the current slicer state. The published screenshot is taken with the "High Value" segment selected and therefore shows the right-hand column. The figures above were recomputed from the source files and match the report.

No drillthrough page, bookmark or report-level filter is configured.

10 Known limitations and caveats

10.1 The data is synthetic

The four Power BI tables are generated. Figures must not be read as findings about any real population.

10.2 The churn score is simulated

The dashboard's churn probability is a simulated column, not the Random Forest output (§2.2). The 0.6 threshold used by `CustomersAtRisk` and by the At Risk segment is therefore a cut-off on a simulated score.

10.3 No time intelligence

A `Date` table exists but is related to Products only, and no visual uses it or `Transactions[Date]`. No month-on-month trend, rolling average or cohort view is available: every figure is a point-in-time snapshot.

10.4 The Products table is loaded but not used

The table is related and two product measures are defined (§8.2), but no visual consumes them. Product penetration is not shown today.

10.5 No row-level security

Every reader sees the whole customer base.

10.6 No revenue measure

The source contains balances, transaction amounts and income, but no fee, margin or interest revenue. Profitability and customer lifetime value cannot be calculated. AnnualIncome is a proxy for customer value.

10.7 Thresholds are hard-coded

The 0.6 risk threshold and the €100,000 / €50,000 segment boundaries are written into the DAX.

11 Opening and reproducing the report

1. Download `MY_Dashboard.pbix` and the four CSV files into a single folder.
2. Open the `.pbix` in Power BI Desktop (Windows). The report opens with its imported data.
3. To refresh, go to Transform data → Data source settings → Change Source and point each of the four queries at the local folder: each query carries its own absolute path from the authoring machine.

To reproduce the model: install `scikit-learn==1.6.1`, pandas, matplotlib and openpyxl, place `Customer-Churn-Records.csv` next to the notebook and run `CHURN MODELING_REPORT.ipynb` from top to bottom. It regenerates `churn_model.pkl` and the prediction files.

12 Roadmap

The following items are not built.

#	Item	Priority	Rationale
1	Replace the simulated churn score with the model output	High	Requires customer data carrying the model's features; closes the gap described in §2.2.
2	Relate the Date table to Transactions and mark it as date table	High	Unlocks transaction trends and time intelligence (§10.3).
3	Promote the three implicit aggregations to named measures	High	Makes every figure traceable to a written definition (§8.3).
4	Calibrate the predicted probabilities	Medium	Isotonic or Platt calibration would make the score a genuine probability (§3.8).
5	Raise recall through tuning and resampling	Medium	Recall of 0.47 is the main operational limit (§3.5).
6	Score the full customer base	Medium	Probabilities are exported for the 2,000-row test set only.
7	Parameterise the risk threshold	Medium	A what-if parameter instead of a hard-coded 0.6.
8	Build the product penetration view	Medium	Table and measures exist; only the visuals are missing (§10.4).
9	Add a separate IsAtRisk flag	Medium	Would surface high-value customers at risk, which the first-match rule hides (§7).
10	Add a customer detail drillthrough page	Medium	From an aggregate risk figure to the customers behind it.
11	Row-level security on relationship manager	Low	Required before sharing with the managers it measures.
12	Host the sources and schedule a refresh	Low	Removes the manual refresh step.

13 Version history

Version	Date	Author	Change
1.0	October 2026	Leprince Dongmo Nanda	First documentation of the delivered solution.
1.1	October 2026	Leprince Dongmo Nanda	Notebook corrected and re-run: the Random Forest pipeline is now the saved model, `RowNumber` removed from the features, single risk-band definition, real `CustomerId` in the export, illustrative hard-coded metrics removed. Model figures updated (§3). Origin of the dashboard churn score, relationships, measure list, data dictionary and Power Query steps aligned with the `.pbix` and the source files.